

Session 6 – Preparation / Integration of Spatial Data (Spatial Operation and Transformation with QGIS)

Objectives: To gain familiarity in a range of spatial operations through QGIS functionalities.

Task 1: Basic Practice to edit Spatial Features

Practice following editing functionalities:

Layer -> Toggling Editing -> Edit Geometry

- > Add Feature, Move Feature, Delete Selected
- > Rotate Feature, Scale Feature, Simplify Feature
- > Reshape Feature, Split Feature
- > Add Ring, Fill Ring, Delete Ring
- > Add Part, Split Part, Delete Part
- > Vertex Tool (have the feature selected first)

Task 2: Segregating and Aggregating Spatial Study Areas

Problem solving: Segregate / aggregate spatial study areas so that data analysis could be carried out on different geographic scales, which allows local characteristics to be taken into account.

Functionality: Boolean Operations on Vector Data – Overlay

Data set: Map of Postcode Areas around London, Map of Inner–Outer London

Add “postarea.shp” and “london_io.shp”, practice the following functionality.

Vector -> Geoprocessing Tools -> Intersect (AND)

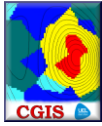
(input layer as “postarea.shp”,
intersect layer as “london_io.shp”)

Postcode Area is large area geography. For detail analysis, it is necessary to segregate it into relative smaller area geography. With Intersect, Postcode Areas are segregated by Inner London region, Outer London region and region outside of London. Please look at the example of Postcode Area of “E”.

Vector -> Geoprocessing Tools -> Dissolve (OR)

(input layer as “postarea.shp”,
Dissolve field as ‘S_N’)

In contrast to segregation, for overarching analysis, Postcode Area can be aggregated by common characteristics. In this case of Dissolve, Postcode Areas are aggregated by South of Thames and North of Thames.



Vector -> Geoprocessing Tools -> Difference (NOT)

(input layer as *"postarea.shp"*,
difference layer as *"london_io.shp"*)

If only Postcode Areas outside of London need to be analysed, simply cut the map of Postcode Areas by London boundary with Difference.

Task 3: Creating Spatial Study Areas

Problem solving: Create new spatial study areas (or zone systems) in order to carry out analysis, simulation or impact assessment on such study areas.

Functionality: Geoprocessing on Vector Data

Data set: Map of credit unions, Map of motorways, Map of London boroughs

Add "credit_union.shp" and "motorway.shp", practice the following functionalities:

Vector -> Geoprocessing Tools -> Convex Hull

(Processing -> Toolbox > *search* "Minimum bounding geometry")

-> Buffer

With Convex Hull and Buffer (distance=3000m), a study area can be created, which covers all locations of credit unions.

With Buffer (distance=1000m), a buffer zone along motorways can be created where environmental impact or economic impact can be assessed.

On "postarea.shp", practice the following functionality.

Vector -> Geometry Tools -> Centroid

With Polygon Centroid, create central points of Postcode Areas. These centroids are useful to represent each Postcode Areas or create a new study area (for example, with Convex Hull and Buffer) for specified data analysis.

These centroids could also help to reduce the computation load, enhance the spatial query and increase the compatibility.

On "credit_union.shp", practice the following functionalities:

Vector -> Geometry Tools -> Delaunay triangulation

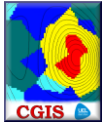
With Delaunay Triangulation, social / economic landscapes could be presented, and impact areas could be created for further studies.

Delaunay Triangulation could also be used to present natural / physical landscape, and for further interpolation.

Create a catchment area using the following functionality

Vector -> Geometry Tools -> Voronoi Polygons (10% buffer region)

With Voronoi Polygons, catchment areas can be created around each credit union. In the same way, GP catchment areas or school catchment areas can be created. On such catchment areas, demand and supply could be analysed.



On “credit_union.shp” and “london_boroughs.shp”, practice following functionality.

Vector -> Data Management Tools -> Join Attributes by Location

(Target layer – credit_union,

Joint layer: london_boroughs)

With Join Attributes by Location, local characteristics can be extracted and assigned on point events. In this case, information of local borough can be assigned on each Credit Union for further analysis (e.g. marketing analysis).

Task 4: Preparing Remote Sensing Data

Problem solving: Extract height information of all features from remote sensing data

Functionality: Calculation on Raster Data

Data set: LiDAR data of DTM and DSM around Ravenscourt Park, Hammersmith, London

Add “TQ2278_DTM_2M.asc” and “TQ2278_DSM_2M.asc”, practice following functionality.

Raster -> Raster Calculator

In this case, heights of all features are extracted from (DSM – DTM). These heights could be further processed / calculated to derive more information, such as roof slope.

Task 5: Working with Data on Different Formats

Problem solving: Extract building height using vector data and raster data

Functionality: Conversion of vector data and raster data

Data set: Map of Building polygon and LiDAR data of DSM around Ravenscourt Park, Hammersmith, London

Add “tq2278_2m.shp”, practice following functionality.

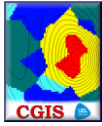
Raster -> Conversion -> Rasterise (Vector to raster)

Convert Building polygons to building raster data so that Building data can be used with DSM. In this case, if DSM data are multiplied with Building raster data, height values of Building can be extracted from all geographic features.

On “TQ2278_DSM_2M.asc”, practice following functionality.

Raster -> Conversion -> Polygonise (Raster to vector)

DSM data can also be converted to polygons which have heights as attributes and can therefore work with other vector data



Task 6: Handling Non-Spatial Attributes

Problem solving: Create two new attribute variables – population density and Household size

Functionality: Attribute Table

Data set: Map of London boroughs

On “london_boroughs.shp”, practice following functionality.

Layer -> Open Attribute Table -> Toggle editing mode

-> Select features using expression

With Attribute Table, create new variables, calculate / update new variables of population density and Household size

Please export the map of Intersect (AND) from **Task 2**, Voronoi Polygon from **Task 3**, height image from **Task 4** (Project -> Import/Export -> Export Map to Image), a CSV table from **Task 6** (Save the vector layer of *London_boroughs* as CSV), and put them into your portfolio.

Because QGIS is an open source software, to which anyone can make contribution. Do you have any suggestion to improve these functionalities you practiced in this session?